

Lecture 24: Seeding social change

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Sociology 204: Social Networks
Princeton University

Wednesday, April 23, 2025



APRIL 2025

SUN	MON	TUE	WED	THU	FRI	SAT
30	31	1	2 Assignment 5 (design) due 11am	3	4 Submit revised design 11:59pm	5 Begin experiment
6	7	8	9 Assignment 6 (progress) due 11am	10	11	12
13	14	15 End experiment	16 Assignment 7 (results) due 11am	17	18	19
20	21	22	23 Assignment 8 (reflection) due 11am	24	25 End of semester	26
27	28	29	30	1	2	3

Logistics

- ▶ there will be no change to submit corrections on the final as you did after the midterm

Social contagion:

- ▶ Obesity is contagious . . . maybe
- ▶ Experimental studies of contagion
- ▶ Complex contagion
- ▶ Seeding social change

Summary

- ▶ differentiate between simple and complex contagions (require 2 or more neighbors to be active)

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- ▶ changes in network structure that promote simple contagion don't always promote complex contagion
- ▶ experiments showing that some behaviors spread more and faster in highly clustered networks rather than random networks

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- ▶ Complex contagion
- ▶ Seeding social change

1. Paluck et al. (2016) Changing climates of conflict: A social network driven experiment in 56 schools. *Proceedings of the National Academy of Sciences*.
2. Airolidi and Christakis (2024). Induction of Social Contagion Across Diverse Outcomes in Structured Experiments in Isolated Honduras Villages. *Science*.

Changing climates of conflict: A social network experiment in 56 schools

Elizabeth Levy Paluck^{a,1}, Hana Shepherd^b, and Peter M. Aronow^{c,d}

Induction of social contagion for diverse outcomes in structured experiments in isolated villages

Edoardo M. Airoidi^{1,2} and Nicholas A. Christakis^{3,4,5*}

Compare and contrast

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- ▶ Settings: Schools in NJ vs Villages in Honduras
- ▶ Thing spreading: Anti-bullying norms vs 117 outcomes rooted in maternal and child health intervention
- ▶ Seed selection strategy: Random vs (Random and Friendship nominated)
- ▶ Network: Spend time with vs spend time with + discuss personal matters + close friends

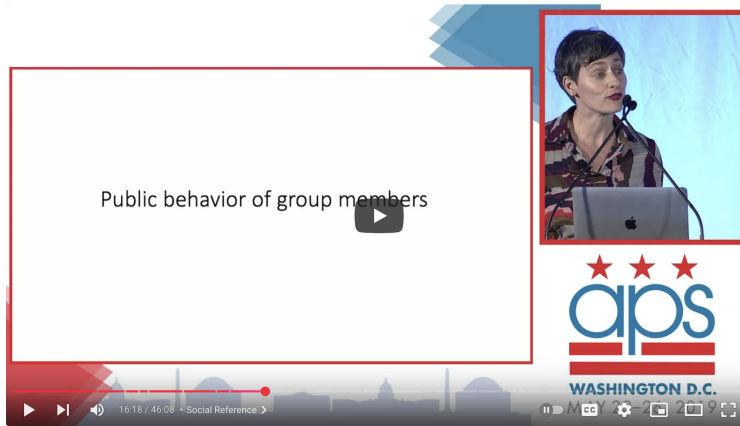
Changing climates of conflict: A social network experiment in 56 schools

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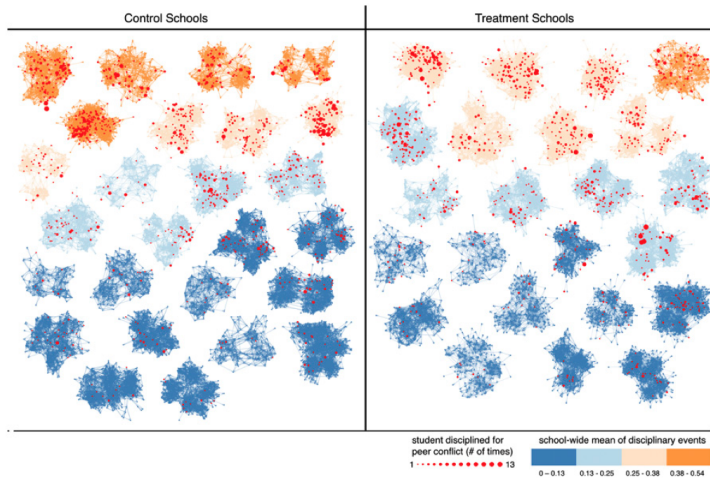
Design:

- ▶ 28 intervention schools and 28 control schools
- ▶ ~30 randomly sampled students

Natural variation in the proportion that are “social referents” (top 10% of indegree)



Let's listen to Betsy describe the experiment and results (about 4 minutes)
<https://youtu.be/DCw6K91xnJ4?si=3aLEgxVuMDenseHY&t=978>



Treatment schools had decrease in disciplinary events by about 30%

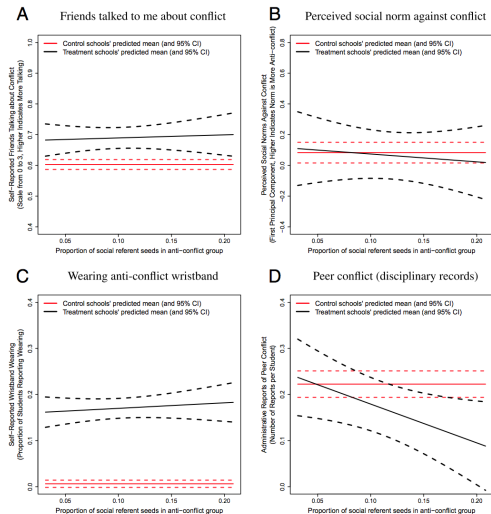


Fig 2D suggests social referents are important but mechanism is not clear from other figures

- ▶ “Whom did you choose to spend time with, face to face or online” (not the same as discuss important matters)

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- ▶ They measured this rather than friendship because it better matched their theory about attention
- ▶ The key feature of the social referent is that their behavior is observed by many other individuals

Why didn't they target social referents?

Why didn't they target social referents? I think they wanted to study the effect no assume it.

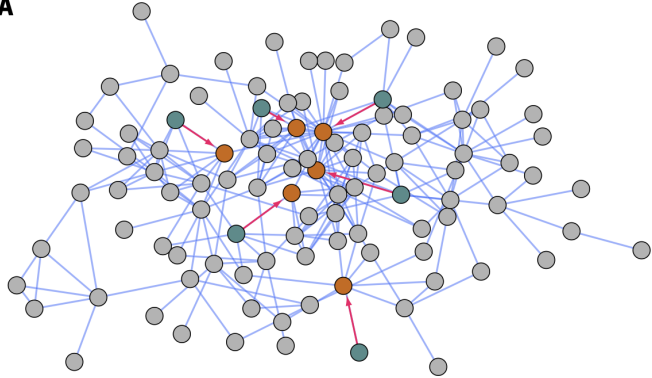
Induction of social contagion for diverse outcomes in structured experiments in isolated villages

Edoardo M. Airoldi^{1,2} and Nicholas A. Christakis^{3,4,5*}

Fig. 1. Conceptual approach to friendship paradox nomination targeting. (A)

Illustrative network map of a hypothetical village of 105 households. Households selected at random (light blue) and households selected by the friendship-nomination technique, whereby a social contact of the random person is in turn randomly chosen (orange), are shown. On average, households selected by friendship nomination are more central in the network and have higher degree compared with the randomly selected households.

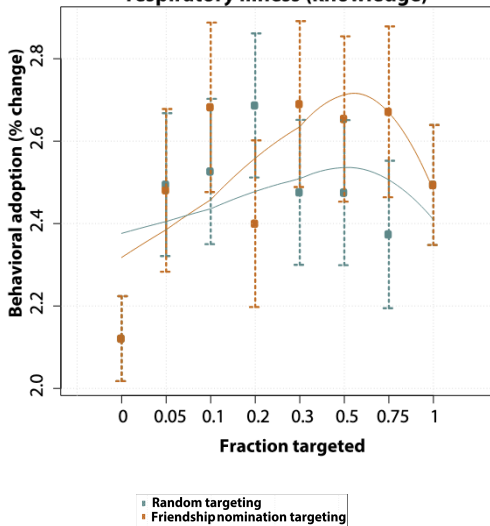
A

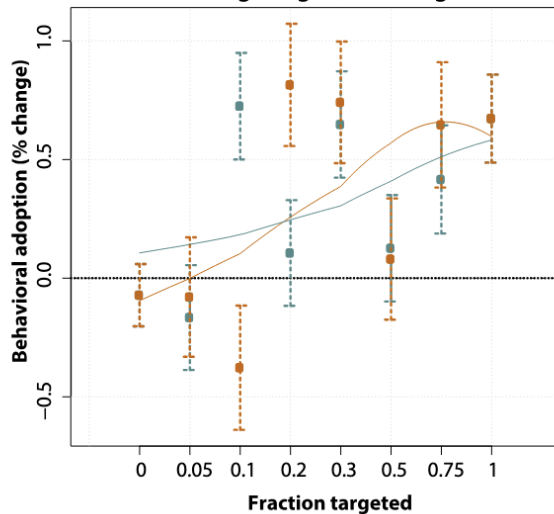


A random sample of friends is different than a random sample of people

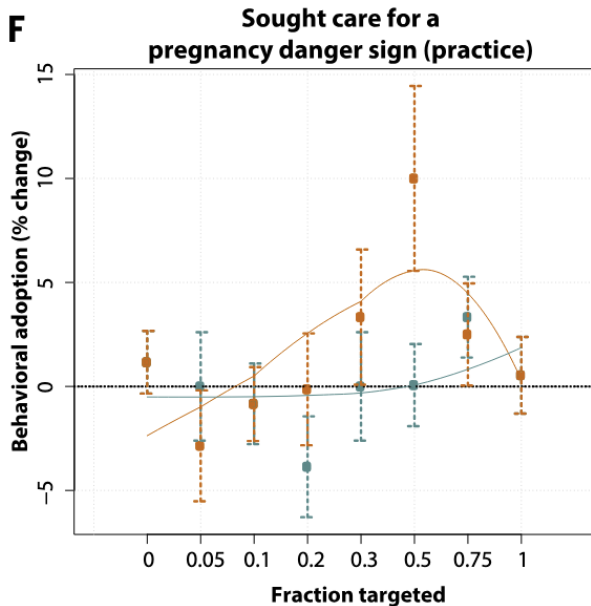
The 176 villages were randomized using an 8×2 factorial design with two axes: (i) the percentage of people targeted per village (0, 5, 10, 20, 30, 50, 75, and 100%) and (ii) the method of choosing the target households (random targeting or friendship paradox-based nomination targeting). Our protocol was published

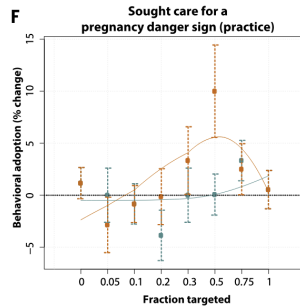
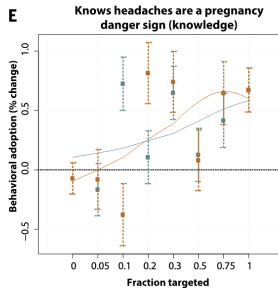
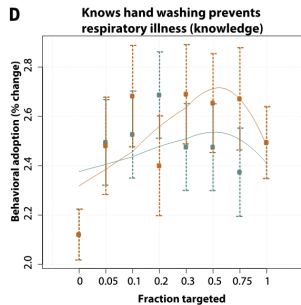
measured 117 outcomes (knowledge, attitudes, and practices)

D**Knows hand washing prevents
respiratory illness (knowledge)**

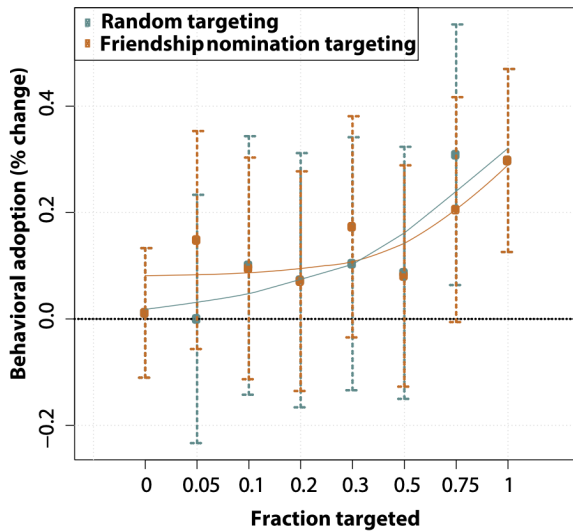
F**Knows headaches are a pregnancy
danger sign (knowledge)**

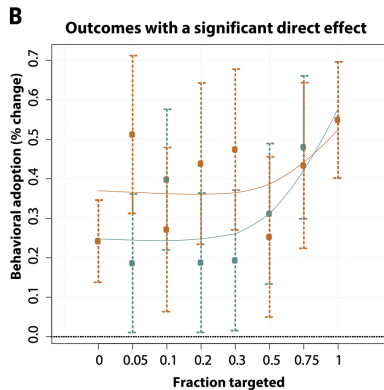
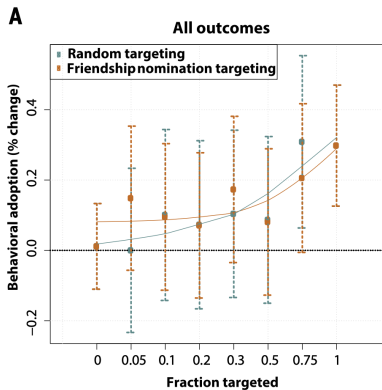
Random targeting
Friendship nomination targeting





■ Random targeting
■ Friendship nomination targeting

A**All outcomes**



The pattern is not super clean; there is probably lots of other stuff happening.

in the 100%-treated villages. Looking at all 117 outcomes, and based on an intention-to-treat analysis, friendship-nomination targeting is discernably more efficient compared with random targeting for 34 of them, with an average reduction in the targeted fraction of households that need to be treated of 7.4% [$P < 0.01$; 95% confidence interval (CI), 5.1 to 9.7%] to obtain an induced change in knowledge, attitudes, or practices that is statistically indistinguishable from selecting households at random

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But did anyone see anything about cost?

Social networks amplify the information and behaviors with which they are seeded (40), but the extent and nature of this effect may depend on the innovation being transmitted. For instance, the intrinsic complexity of understanding or implementing a new behavior, the visibility of its results, its perceived advantage over existing methods, and its baseline prevalence can all influence adoption patterns. Straightforward information (such as knowledge about pregnancy danger signs) might spread by so-called **simple contagion**, requiring only a single contact for transmission between two individuals. Deeper behavioral changes (such as going to a health facility to deliver a baby), by contrast, might require reinforcement from multiple social contacts (e.g., through **complex contagion**) (41), perhaps because they require significant motivation or because they require changes in long-standing beliefs and practices.

Using ideas from this class to help make sense of the findings

Summary:

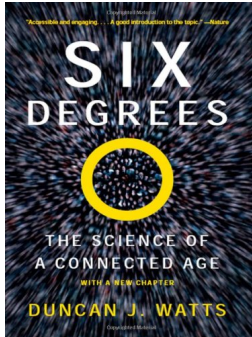
- ▶ a small group of seeds can trigger bigger change

Summary:

- ▶ a small group of seeds can trigger bigger change
- ▶ the way these seeds are selected can impact how much impact they create

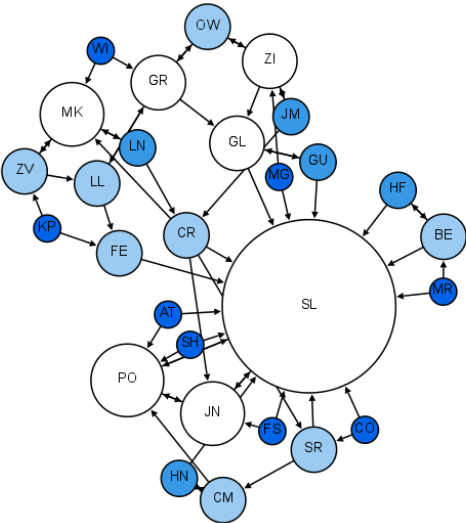
Summary:

- ▶ a small group of seeds can trigger bigger change
- ▶ the way these seeds are selected can impact how much impact they create
- ▶ some things (knowledge) seem easier to spread than others (behavior), which may map on the relationship between simple and complex contagion

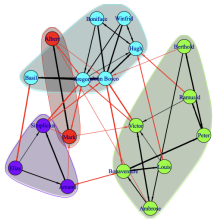


We live in the connected age.

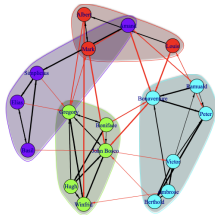
Second grade class, Moreno



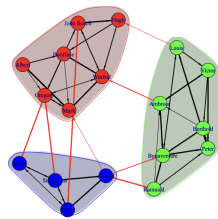
Monks in a monastery, Sampson



Sampson Monks Time 1

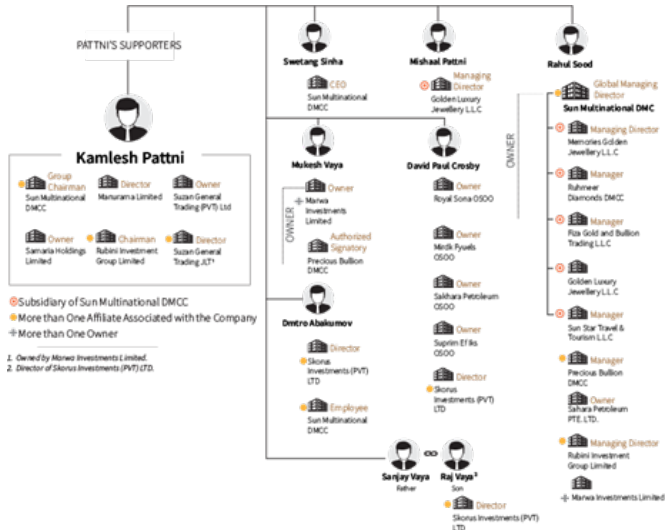


Sampson Monks Time 2

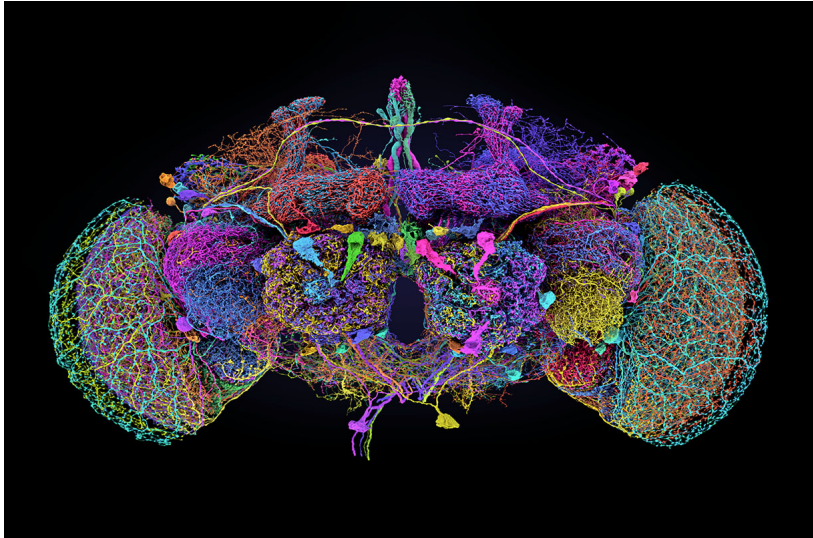


Sampson Monks Time 3

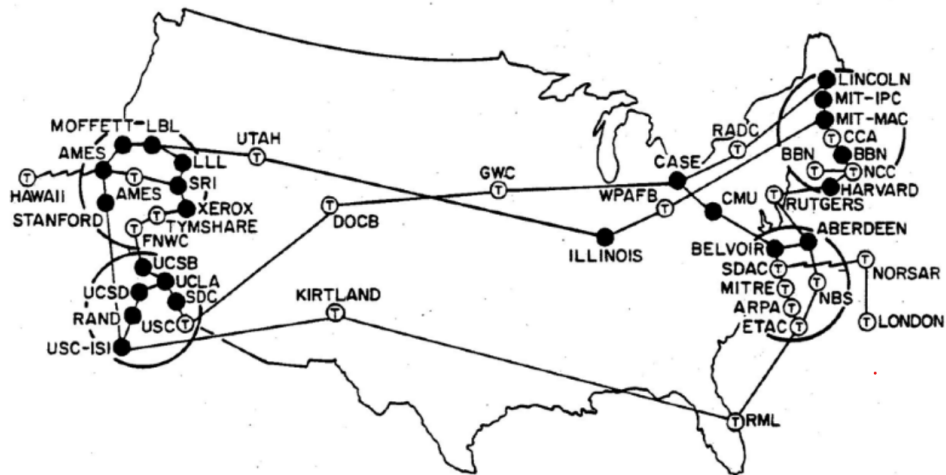
Gold smuggling network of Kamlesh Pattni according to OFAC



Connectome *Drosophila melanogaster* (fruit fly)



ARPAnet in the 1970s



General pattern:

- ▶ nodes
- ▶ edges

Your turn

Think-pair-share: What are some other examples of networks? They don't need to be social.

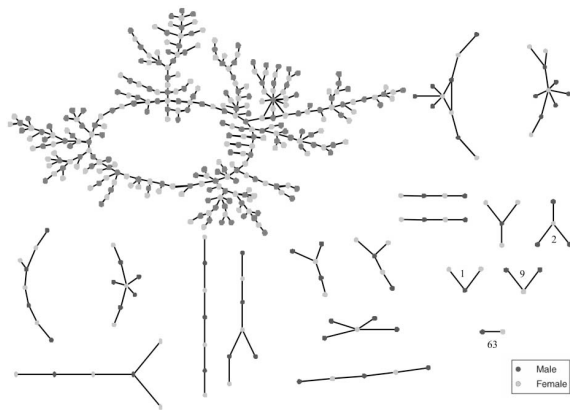


FIG. 2.—The direct relationship structure at Jefferson High

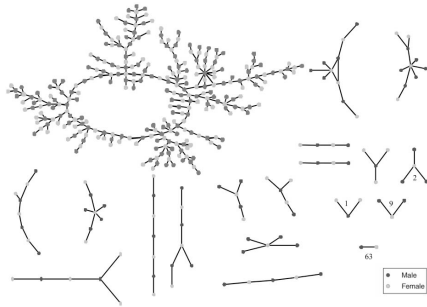
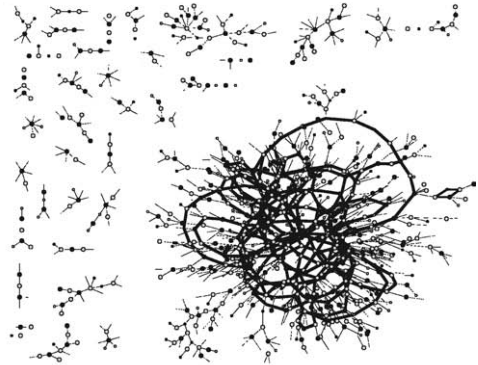


FIG. 2.—The direct relationship structure at Jefferson High

(a) American High School



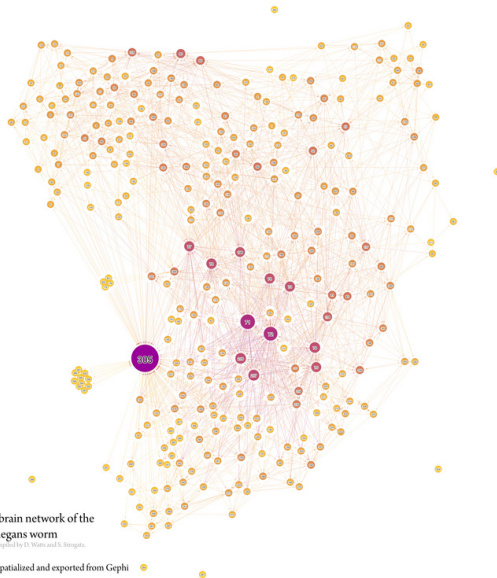
(b) Likoma Island, Malawi

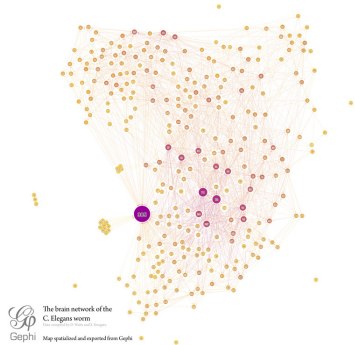


The brain network of the
C. Elegans worm

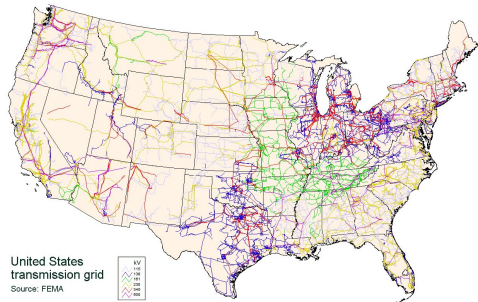
Data compiled by D. Watts and S. Strogatz.

Map spatialized and exported from Gephi





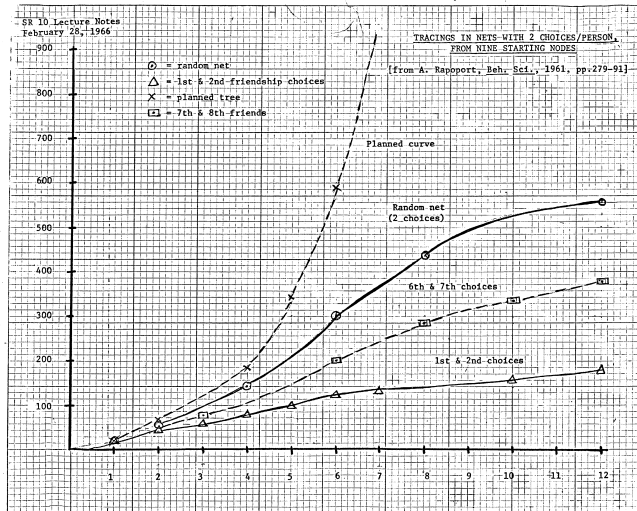
(a) Worm Brain



(b) Power Grid

<https://commons.wikimedia.org/wiki/File:C.elegans-brain-network.jpg> & <https://commons.wikimedia.org/wiki/File:UnitedStatesPowerGrid.jpg>

SR 10 Lecture Notes
February 28, 1966



THE CHAMPION LINE NO. 810
CROSS SECTION - 10 SQUARES TO INCH

Learning objectives

- ▶ Students will be able to **describe** foundational ideas used in the study of networks.

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- ▶ Students will be able to **evaluate** research that applies these ideas to real-world phenomena.
- ▶ Students will be able to **create** research about real-world phenomena.

Thank you for a wonderful semester!